

06-Agent-Architecture.md

Healthcare Clinical Intelligence Platform

Agent Architecture

Version: 1.0

Status: Approved Design

Mission

Build a healthcare-grade multi-agent system that:

- Maximizes answer quality
 - Minimizes hallucinations
 - Uses retrieval efficiently
 - Maintains governance
 - Supports human oversight
 - Minimizes LLM costs
 - Provides full observability
-

Agent Philosophy

Bad:

```
User
↓
LLM
↓
Answer
```

Good:

```
User
↓
Planner Agent
↓
```

```
Retriever System
↓
Verifier Agent
↓
Safety Agent
↓
Response Agent
↓
Answer
```

LangGraph Architecture

```
User Query
|
Planner Agent
|
Retrieval Tool
|
Evidence Validator
|
Confidence Engine
|
Safety Agent
|
Response Agent
|
Final Response
```

Agent Design Principles

Principle 1

Agents do not own knowledge.

Knowledge comes from retrieval.

Principle 2

Agents cannot bypass governance.

Principle 3

Agents cannot access unauthorized data.

Principle 4

Every decision must be traceable.

Agent 1

Planner Agent

Responsibility

Understand user intent.

Determine:

- Query type
 - Risk level
 - Retrieval strategy
 - Required tools
-

Inputs

User Query

User Role

Department

Organization

Conversation Context

Memory Context

Outputs

```
{
  "query_type": "clinical",
  "risk": "high",
  "retrieval_strategy": "hybrid",
  "sources": ["hospital", "pubmed", "fda"]
}
```

Tool Access

Allowed:

- Query Classifier
- Query Rewriter
- Memory Reader

Not Allowed:

- Retrieval
- Final Response

Agent 2

Retrieval Orchestrator

Responsibility

Gather evidence.

Not reasoning.

Calls

- Dense Retrieval
- BM25
- GraphRAG
- Metadata Retrieval

- External Sources
-

Output

Evidence Package

```
{
  "chunks": [],
  "sources": [],
  "confidence": 0.88
}
```

Agent 3

Evidence Validator

Responsibility

Validate retrieval quality.

Checks

Coverage

Freshness

Citation Availability

Source Diversity

Contradictions

Missing Evidence

Output

```
{
  "valid":true,
  "confidence":0.91
}
```

Self Correcting Logic

If:

Confidence < Threshold

Trigger:

```
Query Rewrite
↓
Additional Retrieval
↓
Revalidation
```

Maximum:

3 Attempts

Agent 4

Safety Agent

Responsibility

Healthcare risk management.

Detects

Emergency Symptoms

Pediatric Cases

Pregnancy Cases

High-Risk Drugs

Contraindications

Potential Harm

Example

User:

Chest pain + shortness of breath

Output

Potential emergency.
Seek immediate medical attention.

Override Authority

Can override any response.

Agent 5

Response Agent

Responsibility

Generate final answer.

Never retrieve.

Never override safety.

Inputs

Validated Evidence

Confidence Score

User Role

Memory Context

Outputs

Doctor Format

Research Format

Patient Format

JSON Format

Agent 6

Citation Agent

Responsibility

Build citations.

Generates

Source References

Document Links

Page References

Version References

Agent 7

Memory Agent

Responsibility

Manage memory.

Reads

User Memory

Department Memory

Organization Memory

Approved Agent Memory

Writes

Memory Candidates

Not production memory.

Memory Governance

Agent writes:

Memory Candidate

Not:

Production Memory

Requires approval.

Human Approval Layer

Required For

Knowledge Changes

Memory Promotion

Guideline Updates

Document Approval

Agent Communication Model

All agents communicate through structured state.

Example

```
{
  "query": "",
  "retrieval_results": [],
  "confidence": 0.92,
  "risk": "high"
}
```

No free-form communication.

Agent State

Contains

User Context

Organization Context

Department Context

Memory Context

Retrieval Context

Evaluation Context

Tool Architecture

Retrieval Tool

Used By

Planner

Retriever

Validator

Graph Tool

Used By

Retriever

Memory Tool

Used By

Planner

Memory Agent

Response Agent

Citation Tool

Used By

Citation Agent

Evaluation Tool

Used By

Validator

Cost Optimization

Planner selects path.

Cheap Path

Simple SOP Questions

```
Planner
↓
Retrieval
↓
Response
```

Medium Path

Clinical Questions

```
Planner
↓
Retrieval
↓
Validation
```

↓
Response

Expensive Path

Research Questions

Planner
↓
Retrieval
↓
Validation
↓
Correction
↓
Safety
↓
Response

Agent Failure Recovery

Planner Failure

↓

Fallback Planner

Graph Failure

↓

Hybrid Retrieval

External Source Failure

↓

Cached Sources

No single point of failure.

Observability

Track:

Planner Decisions

Retrieval Strategy

Retrieved Sources

Validation Results

Safety Flags

Memory Reads

Memory Writes

Tool Calls

Cost

Latency

Evaluation Metrics

Planner Accuracy

Tool Selection Accuracy

Retrieval Success Rate

Validation Success Rate

Safety Detection Rate

Citation Accuracy

Answer Quality

Cost Per Query

Latency

LangGraph Nodes

Query Intake

Planner

Memory Loader

Retrieval Orchestrator

Evidence Validator

Confidence Engine

Corrective Retrieval

Safety Agent

Citation Agent

Response Agent

Audit Logger

Feedback Collector

Future Agents

Research Agent

Workflow Agent

Insurance Agent

Clinical Trial Agent

Final Principle

Agents should coordinate reasoning.

Retrieval should provide knowledge.

Governance should provide trust.

Safety should always have final authority.